PULSE

Hypertension monitoring, early warning and emergency support

Final conception, draft for team critique

36-hour hackathon · 3-person team · version 0.9 · 29 October 2026

Status: NOT approved. No code has been written. Approve or amend before production starts.

What this document is

The single reference the team reviews before writing code. It turns the hackathon brief, the first team concept and the conception handoff into concrete decisions: scope, emergency workflow, architecture, repository and merge workflow, data model, API contract, alert rules, tests and a 36-hour plan.

How to review it

- Read sections 1 to 4 together (about 15 minutes). They decide what PULSE is.
- Each member then reviews the sections they own (see the reading map below) and writes comments.
- Settle every item in section 19 *Open decisions*. Mark each as approved, changed or rejected.
- Ayoub records the outcome in `docs/decisions.md` in the repo. That file, plus the frozen API contract, is what production follows.

Reader	Must read	Owns
Ayoub (backend + merge lead)	All	Sections 6, 8 to 12, 15.2, 17, 19. Final say on contract and merges.
Frontend developer	1 to 7, 11, 14 to 17	Section 14 and the screen behaviour in section 7.
Data / AI developer	1 to 6, 10, 11, 13, 15 to 17	Sections 5, 13 and the test vectors in 15.1.

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1. The concept in one page

PULSE is a mobile-first web app for people with hypertension and the doctors who follow them. Patients record blood pressure readings (typed in, or from a clearly labelled simulated device). PULSE explains each reading against a clinician-set target, spots worrying trends, and when a reading is dangerously high or the patient reports a red-flag symptom, it switches to a short emergency flow: call the emergency number, show a medical card a bystander can read, show where the patient is, and reach a trusted family member. Every alert and emergency is saved as an incident the patient's own doctor reviews later from the doctor dashboard, where they leave a follow-up note.

> **Product promise (use this sentence in the pitch)**
>
> PULSE helps patients and the people around them know the next step when blood pressure or symptoms become dangerous, and keeps a clear record so the patient's own doctor can follow up.

Priority order (when time runs out, cut from the bottom)

#	Priority	Why
1	Emergency action is clear, fast and reliable	The team's stated goal: help the patient survive a possible emergency.
2	Readings and alerts are handled correctly	Wrong alerts are worse than none. Rules must be deterministic and tested.
3	Doctor follow-up and history	Answers the official problem: visibility between consultations.
4	Journal, lifestyle guidance, AI summary	Valuable, but only after 1 to 3 work end to end.

Three paths that must never be confused

Path	Who	When	What PULSE does
Emergency services	Protection Civile / SAMU via the phone dialer	Red-flag symptoms, or severe reading with symptoms	Big call button to a configurable, team-verified number. Shown first, never behind a form or a map.
Trusted contact	Family member chosen by the patient	Any emergency, after or alongside calling	Call button and a prepared share message with location. Supplements, never replaces, the emergency call.
Regular doctor	The patient's own doctor in PULSE	Afterwards, for follow-up	Sees the incident, readings before and after, and journal. Writes a note. Not notified in real time.

> **Key correction to the first concept**
>
> The patient's own clinic is not where an emergency goes. In an emergency PULSE points to emergency services. The regular doctor is for follow-up, through the incident record.

2. What changed from the first team concept

Everything from the first concept is kept unless it was unsafe, impossible in a web app, or not buildable in 36 hours. Each change below is open to critique.

First concept	Decision	Reason
Loud sound to alert people around a patient who passed out	Kept as an optional 'Attract attention' button on the emergency screen. Not the main feature.	A browser cannot play sound without a tap, cannot run in the background, and cannot detect fainting. A sound alone summons nobody.
Detect that the patient passed out	Out of scope.	A blood pressure cuff does not detect consciousness. No validated sensor is available.
Emergency route to own clinic / private doctor / nearest hospital	Emergency route goes to emergency services. Own doctor gets the incident later. Nearest hospital is an informational map link only (if time).	The nearest or usual clinic may have no emergency department. A map search confirms nothing about capacity.
Send message to ambulance with location	Out of scope. The patient or bystander calls; PULSE shows the location to read out.	No public dispatch API in Algeria that a hackathon can integrate with legitimately.
Alert to family member	Kept: call button + prepared share message with location.	Works through the phone's own share sheet. PULSE never claims the message was sent.
Live sensor data	Manual entry + simulated device feed labelled 'simulated'. Real device only if one is already on hand and tested.	Bluetooth cuffs need Web Bluetooth (not on iPhone) and per-model work.
Sport and food: what to do / avoid this week	If time permits, as general education cards chosen by weekly status band. Doctor's own advice shown separately.	Personalised diet or exercise prescriptions are medical advice. General guidance is safe if sourced and reviewed.
Weekly consultation / journal	MVP: short entries with 'ask my doctor' flag; doctor marks reviewed and can reply.	Cheap to build and directly supports doctor follow-up.
Two dashboards (patient, doctor)	Kept. One app, two roles, one database.	Both views must show the same persisted data.

3. Users and what each one needs

User	Goal	Key screens
Patient	Record readings, understand them, get help fast when needed, ask the doctor questions.	Home, Add reading, History, Alerts, Emergency, Journal, Profile
Bystander	Understand within seconds what to do and whom to call, without knowing the app.	Emergency screen (no login wall once the phone is unlocked), lock-screen card / native Medical ID
Trusted contact	Know that the patient needs help and where they are.	No account in the MVP. Receives a phone call or a shared message.
Doctor	See which assigned patients need attention, review incidents, leave follow-up.	Patient list, Patient detail, Alert review, Incident report, Notes

4. Scope

Feature	Status	Owner	Notes
Demo login for patient and doctor roles, access restricted by assignment	MVP	Back	Seeded accounts. No registration screen.
Patient profile: conditions, medications (as entered), allergies, home address, regular doctor, trusted contacts	MVP	Back + Front	Patient edits; doctor reads.
Add reading (manual) with context (rested 5 min, position)	MVP	Front	Validation in analysis module.
Simulated device readings, labelled 'simulated' everywhere	MVP	Data	Script + dev endpoint.
Reading history + trend chart + 7-day summary	MVP	Front + Data	Chart reads the same API data.
Deterministic alert rules with reason text	MVP	Data	Section 5.
Symptom check after a severe reading; red-flag symptoms trigger emergency	MVP	Data + Front	Also reachable directly: 'I feel unwell'.
Recheck timer (60 s) after a severe reading	MVP	Front	Skipped when symptoms are present.
Alert state machine with audit trail	MVP	Back	Section 6.
Emergency screen: call, medical card, location, manual place, trusted contact	MVP	Front	Works offline from local cache.
Incident record + action log + patient-reported outcome	MVP	Back + Front	Logs taps, never claims success.
Doctor: patient list by urgency, patient detail, alert review, incident report, notes, care-plan target	MVP	Front + Back	
Journal with 'ask my doctor', doctor review and reply	MVP	Front + Back	Minimal form and list.
Emergency screen text in French and Arabic	MVP (static text)	Front	Rest of UI French. Bystanders may read either.
Demo mode: emergency number replaced by a team phone, visible banner	MVP	Back + Front	Never dial real emergency services in a demo.
Lock-screen emergency card image generator	If time (high)	Front	Patient picks fields; saves PNG to set as wallpaper.
Native Medical ID setup guide (Android / iPhone)	If time (high)	Front	Static steps + copyable text from profile.
'Attract attention' sound button	If time	Front	Tap-triggered only.
Weekly general guidance cards (activity, salt, sleep)	If time	Data	Sourced, reviewed, labelled general.
AI-drafted weekly summary for the doctor	If time	Data/AI	Doctor-only, labelled, from computed stats. Section 13.
'Find emergency departments near me' map link	If time	Front	Opens a map search. Labelled informational.
Missing-reading reminder (in-app banner)	If time	Data + Front	No push notifications.
Arabic UI throughout (RTL)	Out		RTL across all screens costs too much time.
Ambulance dispatch, SMS gateway, doctor push/SMS notification	Out		Not built; never implied.
Fainting / unconsciousness detection, background monitoring	Out		Not possible in a web app; no sensor.

Feature	Status	Owner	Notes
Diagnosis, medication advice, AI talking to the patient	Out		Safety boundary.
EMR / hospital integration, real patient data	Out		Fake demo data only.

5. Alert logic and medical-safety boundaries

Safety boundary

- Rules are a prototype policy taken from public American Heart Association patient guidance. A qualified clinician must review thresholds and wording before any real use.
- PULSE never diagnoses, never tells anyone to start, stop or change medication, and never says a reading is 'fine' in a way that discourages seeking care.
- Every alert shows: the reading, the rule in plain words, the time, and one clear next step.

5.1 Reading validation (runs before any rule)

ID	Check	Result
V1	Systolic 50 to 300, diastolic 30 to 200, pulse 25 to 250 (if given), integers	Reject with 422 and a field message
V2	Systolic must be greater than diastolic	Reject with 422
V3	measured_at not more than 5 minutes in the future	Reject with 422
V4	Pulse pressure (systolic minus diastolic) below 20 or above 120	Store as `questionable`, still evaluated, suggest re-measuring

A questionable reading is never silently changed or dropped. A possibly severe reading is still treated as severe.

5.2 Red-flag symptoms (checklist shown to the patient)

From the AHA hypertensive-crisis guidance: **chest pain, shortness of breath, back pain, numbness, weakness, change in vision, difficulty speaking**. Stored as codes: `chest_pain`, `short_breath`, `back_pain`, `numbness`, `weakness`, `vision_change`, `speech_difficulty`. The clinician may extend the list (for example severe headache or confusion).

5.3 Rules

Rule	Condition	Level	Patient sees (next step)	Doctor sees
R-EMG-1	Any red-flag symptom reported, whatever the reading	emergency	Emergency screen immediately. 'Call emergency services now.' No recheck.	Top of list, red, incident linked
R-SEV-1	Systolic ≥ 180 or diastolic ≥ 120	severe	'Very high reading.' Symptom checklist first. If none: rest and re-measure in 60 s (timer).	Red badge, open alert
R-SEV-2	Recheck within 15 min still ≥ 180 / ≥ 120, no symptoms	severe (persistent)	'Still very high. Contact your doctor or seek urgent medical advice today. If symptoms appear, call emergency services.'	Marked persistent
R-SEV-3	Recheck within 15 min below 180 / 120	severe (stays open)	'Lower now. Your doctor will still see this reading.'	Alert stays for review
R-TGT-1	Reading ≥ patient target (default 140/90 until doctor sets one)	notice	'Above your target. Rest 5 minutes and measure again.' Can be dismissed.	Counted in summary

Rule	Condition	Level	Patient sees (next step)	Doctor sees
R-TRD-1	≥ 3 readings in last 7 days and their mean ≥ target	trend	'Your readings this week are above target. Your doctor will review them.'	Amber, needs review
(none)	Below target	no alert	Saved; chart updated.	Chart

De-duplication

- One open `notice` and one open `trend` alert per patient. New matching readings attach to the open alert (event `measurement_attached`) instead of creating more.
- Severe and emergency alerts are never merged away. Each is its own alert.
- Severity can rise on an open alert (severe to emergency when symptoms are reported) and is logged as `level_raised`. It never goes down.

5.4 Wording rules for all patient-facing text

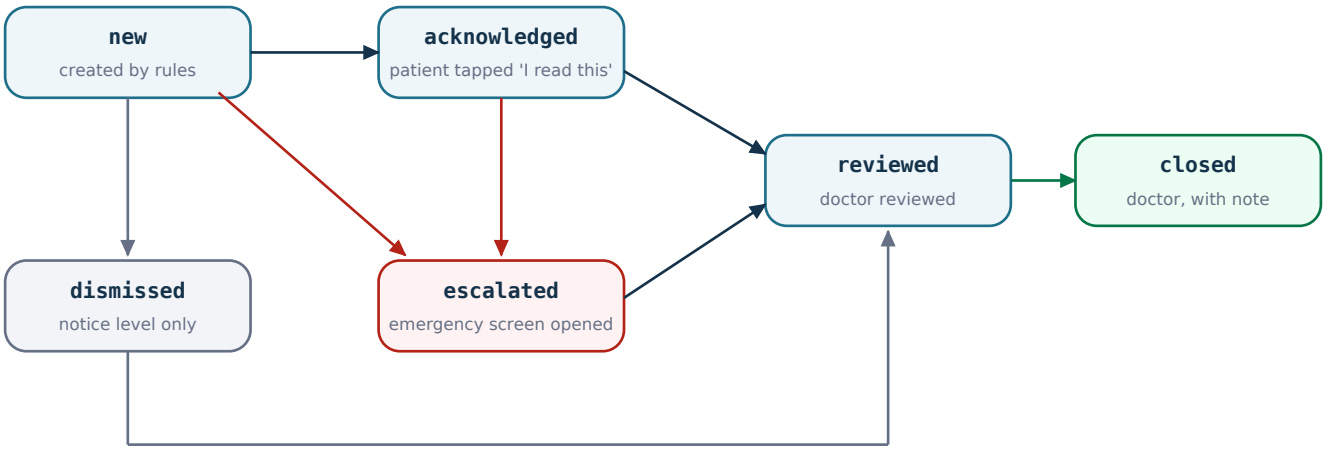
- Plain French, short sentences, one action per message. Emergency screen also in Arabic.
- Say 'your target' only when a doctor set it; otherwise 'the general target'.
- Never: 'you are fine', 'normal, no need to see a doctor', 'take / stop / increase your medicine'.
- Always include, on severe and emergency screens: 'If you feel unwell, call emergency services.'
- Data source always visible next to the number: Manual, Simulated, Device.

5.5 Needs clinician review before real use

Item	Prototype value
Crisis threshold	≥ 180 systolic or ≥ 120 diastolic (AHA uses 180/120 or higher)
Recheck delay	60 s in app (AHA: wait 1 to 2 minutes)
Default target before doctor sets one	140/90 (alternative for home readings: 135/85)
Trend rule window and count	7 days, at least 3 readings, mean ≥ target
Symptom list	AHA 7 symptoms above
Low readings (possible fainting risk)	No rule in MVP. Clinician to decide.
All patient-facing text	French and Arabic strings in one file for review

6. Alert state machine

Any open state can be reviewed by the assigned doctor. Severity can only go up (level_raised event). Nothing is deleted.



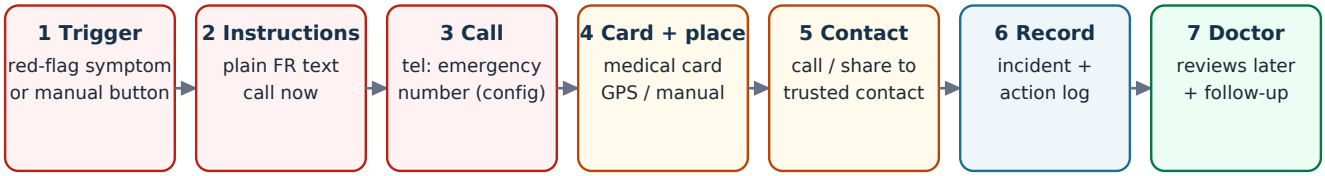
Transition	Who	Allowed for levels	Side effects
new → acknowledged	Patient	all	event logged

Transition	Who	Allowed for levels	Side effects
new / acknowledged → escalated	Patient or system (emergency screen opened from this alert)	severe, emergency	incident created and linked
new / acknowledged → dismissed	Patient	notice only, reason required	doctor still sees it in history
any open → reviewed	Assigned doctor	all	optional note
reviewed → closed	Assigned doctor	all; note required for severe / emergency	doctor_note linked
anything else	-	-	409 Conflict

Every change writes an `alert_events` row (alert, from, to, actor, role, note, time). Opening an alert never changes its state. Nothing is deleted. Transitions are enforced in one backend function covered by tests.

7. The emergency workflow (core of the product)

Red = survival path (must work first) Amber = support Teal/green = record and follow-up



Steps 1-5 render from the phone's local cache and never wait for GPS or network. Step 6 is queued and retried if offline.

7.1 What a web app can and cannot do

Capability	Mobile web app (our MVP)	Native app (future)
Open the dialer with the number filled in	Yes, tel: link. User confirms the call.	Yes
Know the call connected	No. Logged as 'call button tapped'.	Partly
GPS location	Yes, only over HTTPS and after permission. Can time out indoors.	Yes, more reliable
Share location to a contact	Yes via share sheet (navigator.share) or SMS composer. User presses Send.	Can send SMS directly with permission
Work when phone is locked	No	Partly (widgets, lock-screen notifications)
Run in background, monitor continuously	No	Limited by OS
Play a loud sound	Only after a tap, while the page is open	Yes, with permission
Show info on the lock screen	No. Workaround: lock-screen card image and native Medical ID.	Native Medical ID / Emergency info
Work without network	Yes for the emergency screen, if cached at login	Yes

7.2 If the phone is locked or the patient is unconscious

PULSE cannot solve this inside the browser, so it relies on three honest layers:

Layer	How it works	Status
1. Native phone emergency info	Android 'Emergency information' and iPhone 'Medical ID' (with 'Show when locked') are readable from the lock screen's Emergency button. PULSE gives step-by-step setup and a copyable text built from the profile.	If time (high)
2. Lock-screen card image	PULSE generates a PNG with the fields the patient chooses (name, key conditions, allergies, contact phone, 'In emergency call 14'). Patient sets it as lock-screen wallpaper. Works offline, no unlock needed. Privacy: anyone who sees the phone can read it, so the patient chooses the fields.	If time (high)
3. In-app Emergency screen	One red button on every screen. Renders from local cache, no network needed. For a bystander when the phone is unlocked or the patient can unlock it.	MVP

> **Say this in the pitch**
> 'If the phone is locked, a bystander uses the phone's own emergency information, which PULSE helps the patient set up. If it is unlocked, PULSE gives them one button and a card to read out.'

7.3 Emergency screen specification

- Reached by: the red 'Urgence' button in the header of every patient screen, a red-flag symptom, or a severe alert. Maximum one tap from any screen.
- Top: large red button **Appeler les secours** with the number and service name visible. Under it: 'Call first. Then contact family.' in French and Arabic.
- Second: trusted contact card: name, relationship, **Call**, **Share my location**.
- Third: medical card for a responder: name, age, key conditions, medications as entered, allergies, time and value of last reading.
- Fourth: location: coordinates and accuracy, 'Open in Maps' link, or saved home address, plus a text field 'Where am I? (landmark)'. Never blocks the call button.
- Bottom: 'I'm OK now / What happened?' to report outcome; 'This is a prototype' notice; DEMO MODE banner when on.

7.4 Emergency numbers in Algeria

Service	Number	What we found	Status
Protection Civile	14	Widely listed for Civil Protection; the UK FCDO page lists 14 for fire.	Confirm locally
Protection Civile	1021	UK FCDO lists 1021 for ambulance; Wikipedia lists it as Civil Defence.	Confirm locally
Police	17 / 1548	Listed by UK FCDO.	Not used in app
Gendarmerie	1055	Listed on Wikipedia.	Not used in app
SAMU	varies	Local SAMU numbers are reported per wilaya and could not be verified.	Do not use

- Numbers live in `emergency_config` (served by `GET /config/emergency`), each with `verified` and `source`. No number is hard-coded in the frontend.
- Before the demo, confirm the main button's number with the hackathon's medical mentor or the official Protection Civile site, and record who confirmed it and when.
- **Never test-call an emergency number.** Demo mode swaps it for a team member's phone and shows a 'DEMO MODE' banner.

7.5 Location and sharing

- Ask for location permission during onboarding with an explanation, not during the emergency.
- In the emergency: request position with an 8-second timeout. Show coordinates, accuracy and a maps link; otherwise saved home address and the manual field.
- Share message (FR): 'PULSE : [Name] a besoin d'aide. Position : [maps link] ([time]). Dernière tension : 182/121 à 14:05. Appelez-le/la ou appelez le 14.'
- Order of share fallbacks: `navigator.share` → `sms:` link with body → copy to clipboard. Each attempt and failure is logged.
- Logged as `share_opened`, never 'sent'.

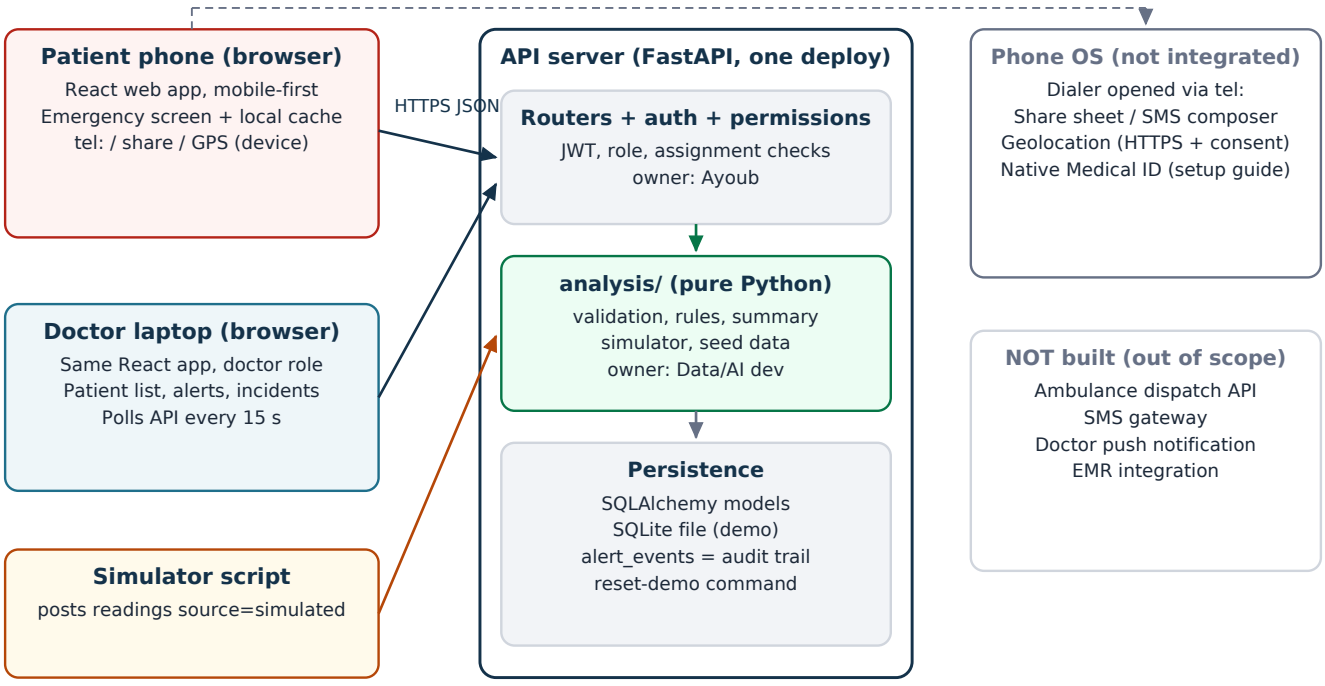
7.6 Incident record and what the doctor sees later

Saved in the incident	Shown to the doctor
Trigger (symptom, severe reading, manual button), start time	Incident card at the top of the patient detail
Linked alert and readings, symptoms reported	Readings from 24 h before to 24 h after on the chart, marked
Location snapshot (only if consented): lat, lng, accuracy, source	'Location shared: yes / no / manual'
Action log: screen opened, call tapped, contact tapped, share opened / failed, location granted / denied / timeout	Timeline with plain labels, e.g. 'Emergency call button tapped (call not confirmed)'
Patient-reported outcome: called emergency, went to hospital, family came, false alarm, unknown	Outcome and patient's note
Doctor note, advice for patient, follow-up date	Written by the doctor; advice part visible to the patient, labelled with the doctor's name

The doctor is not notified in real time

Real-time notification is out of scope. The dashboard sorts patients with open emergencies to the top and refreshes every 15 seconds while open. The pitch must not say 'the doctor is alerted'.

8. Architecture



8.1 Recommended stack (decision D1)

Layer	Choice	Why
Frontend	React 18 + Vite + TypeScript, Tailwind CSS, Recharts, React Router	Fast to build, mobile-first, good chart library.
Backend	Python 3.12, FastAPI, SQLAlchemy 2, Pydantic v2, JWT (python-jose), passlib/bcrypt	Same language as the data/AI developer; Pydantic models are the contract; OpenAPI generated automatically.
Database	SQLite file for the hackathon (Postgres-compatible models)	Zero setup, easy reset. Swap to Postgres later by changing the URL.
Contract	openapi.json generated by FastAPI; TS types generated with openapi-typescript	Frontend types always match backend. No hand-copied shapes.
Tests	pytest (rules + API), Vitest for a few frontend utils, manual demo checklist	Rules are the most important thing to test.
Deploy	One service: FastAPI serves the built frontend. Render or Railway with HTTPS. Backup: laptop + Cloudflare tunnel.	One URL, no CORS, HTTPS needed for GPS on the phone.

If the team prefers Node for the backend, the same design works with Express + Prisma, but the data/AI developer's Python rules would then need porting to TypeScript. That is the main reason for FastAPI.

8.2 Design rules

- One database, one API. Both dashboards read the same records. No separate mock datasets after hour 4.
- The analysis module is pure functions (no database, no HTTP). The backend calls it and saves the result.
- All timestamps stored in UTC (ISO 8601). Display in Africa/Algiers time.
- Emergency card and config are cached in the browser at login and refreshed on each app open, so the emergency screen opens offline.
- Incident actions are queued in the browser and retried until the API accepts them; each carries the client time.
- Doctor dashboard refresh: poll every 15 s. No WebSockets.

9. Repository and team workflow

9.1 Roles

Person	Role	Owns in the repo	Responsible for
Ayoub	Backend + merge lead	api/ (except analysis/), contract/, docs/, CI, deploy	Contract, auth, permissions, persistence, state machine, incidents, reviews and merges every PR, keeps main demo-ready.
Frontend dev	Frontend	web/	All screens, emergency screen behaviour (tel, share, GPS, offline cache), charts, mobile testing.
Data / AI dev	Data + analysis	api/app/analysis/, api/app/seed/, analysis tests	Validation, rules, summary stats, test vectors, seed data, simulator, guidance content, optional AI summary.

9.2 Repository layout (one GitHub repo, monorepo)

```
pulse/
├─ README.md                  how to run, demo accounts, demo mode
├─ docs/
│  ├─ conception.pdf          this document (approved version)
│  ├─ decisions.md            outcome of section 19, dated
│  ├─ demo-script.md
│  └─ clinical-review.md      thresholds + texts for clinician sign-off
├─ contract/
│  └─ openapi.json            generated, committed, reviewed in PRs
├─ api/
│  ├─ pyproject.toml
│  ├─ app/
│  │  ├─ main.py  config.py  db.py  models.py
│  │  ├─ schemas.py           Pydantic request/response models = THE CONTRACT
│  │  ├─ auth.py  permissions.py  state_machine.py
│  │  ├─ routers/  auth.py patients.py measurements.py symptoms.py alerts.py
│  │  │            incidents.py journal.py notes.py config.py dev.py
│  │  ├─ analysis/            (Data/AI dev) pure functions, no DB
│  │  │  ├─ validation.py  rules.py  summary.py  symptoms.py
│  │  │  ├─ simulator.py  guidance.json  ai_summary.py (if time)
│  │  │  └─ texts_fr.json texts_ar.json   all patient-facing alert strings
│  │  └─ seed/demo_seed.py    3 patients, 1 doctor, 30 days of readings
│  └─ tests/  test_rules.py test_validation.py test_summary.py
│             test_api_permissions.py test_api_flow.py test_state_machine.py
├─ web/
│  ├─ package.json  vite.config.ts
│  └─ src/
│     ├─ api/                 generated types + thin fetch client
│     ├─ lib/                 geo.ts share.ts offlineCache.ts actionQueue.ts time.ts
│     ├─ components/          EmergencyButton, ReadingBadge, SourceTag, Chart...
│     └─ pages/patient/  pages/doctor/  pages/emergency/
├─ scripts/  gen-types.sh  reset-demo.sh  simulate.sh
├─ .github/  CODEOWNERS  workflows/ci.yml  pull_request_template.md
└─ Makefile  dev | test | reset-demo | gen-types | build
```

9.3 Branching and merging

- **main** is protected: only Ayoub merges, CI must pass, and main must always run the demo.
- Short-lived branches named by area: `feat/web-emergency-screen`, `feat/data-rules`, `feat/api-incidents`, `fix/...`. One feature per branch, merged within about 3 hours.
- Pull often: everyone rebases on main at each checkpoint (section 17).
- CODEOWNERS: `web/` → frontend dev, `api/app/analysis/` → data dev, everything else → Ayoub. Ayoub approves every merge.
- Commit style: `feat(api): add incident actions`, `test(data): severe recheck vectors`.

9.4 Contract-first rule (prevents integration surprises)

- By hour 3 Ayoub commits `schemas.py`, stub routers returning seeded data, and `contract/openapi.json`. That is the contract freeze.
- After the freeze, any field change goes through a PR that edits `schemas.py`, regenerates `openapi.json` and the TS types, and is announced in the team chat.
- Frontend may use JSON fixtures copied from the contract examples only until the stubs exist (about hour 4). Then it calls the real local API with seeded data.
- The analysis module exposes only the functions in section 13. The backend never re-implements rule logic.

9.5 CI and definition of done

CI step (GitHub Actions)	Must pass to merge
pytest api/tests	Yes
Check openapi.json is up to date (regenerate and diff)	Yes
web: typecheck (tsc --noEmit) and vite build	Yes
ruff / eslint	Warnings allowed after hour 24

Done means: merged into main, works against the real API with seed data, tested on a phone screen width, and listed with its status (implemented / simulated / proposed) in the README.

10. Data model

13 tables. IDs are integers for simplicity. All times UTC. *JSON* means a JSON column (SQLite TEXT).

Table	Columns	Notes
users	id, role (patient | doctor), full_name, email (unique), password_hash, created_at	Seeded only
patient_profiles	user_id (PK, FK users), date_of_birth, sex, conditions JSON, medications JSON, allergies JSON, home_address, regular_doctor_name, regular_clinic_name, regular_clinic_phone, location_consent bool, emergency_card_fields JSON, updated_at	Medications are 'as entered', never advice
doctor_patient_assignments	doctor_id, patient_id, assigned_at (PK doctor_id + patient_id)	Basis of every doctor permission
care_plans	id, patient_id, target_systolic, target_diastolic, note, author_doctor_id, effective_from	Latest one applies; none → default 140/90
measurements	id, patient_id, systolic, diastolic, pulse null, measured_at, received_at, source (manual | simulated | device), context JSON (rested, position, arm), validation_status (valid | questionable), validation_notes, created_by	Never edited after creation
symptom_reports	id, patient_id, symptoms JSON (codes), none_reported bool, notes, reported_at, measurement_id null	
alerts	id, patient_id, level (notice | trend | severe | emergency), rule_id, reason_text, next_step_text, measurement_id null, symptom_report_id null, state, persistent bool, created_at, updated_at	Level only rises
alert_events	id, alert_id, type (state_change | level_raised | measurement_attached | note), from_state, to_state, actor_user_id null, actor_role (patient | doctor | system), note, created_at	Audit trail; append-only
trusted_contacts	id, patient_id, name, relationship, phone, is_primary, may_receive_location bool	No account
incidents	id, patient_id, alert_id null, trigger (symptom | severe_reading | manual), started_at, location_lat, location_lng, location_accuracy_m, location_source (gps | manual | home | none), manual_location_text, outcome null, outcome_note, ended_at	Location only if consent
incident_actions	id, incident_id, action (enum below), detail JSON, client_time, server_time	Append-only
journal_entries	id, patient_id, category (symptom | food | activity | medication | question | other), text, ask_doctor bool, created_at, review_status (pending | reviewed), reviewed_by, reviewed_at	
doctor_notes	id, doctor_id, patient_id, alert_id null, incident_id null, journal_entry_id null, private_note, advice_for_patient, follow_up_date null, created_at	Patient sees advice only

incident_actions.action: `screen_opened, call_emergency_tapped, call_contact_tapped, share_opened, share_failed, location_granted, location_denied, location_timeout, manual_location_entered, sound_played, outcome_reported`.

emergency_config is a settings file (`config.py` + env), not a table: country, demo_mode, demo_number, numbers [label, number, verified, source, confirmed_by, confirmed_on].

11. API contract (v1)

Base path `/api/v1`. JSON. Auth: `Authorization: Bearer <JWT>` (contains user id and role). The full schema is `contract/openapi.json`; this table is the agreed surface.

Method + path	Who	Purpose
POST /auth/login	anyone	email + password → token + user
GET /me	any	current user
GET /config/emergency	any	numbers, demo_mode, disclaimer
GET /patients	doctor	assigned patients with summary, sorted by urgency
GET /patients/{id}/profile · PUT	patient own (PUT) · doctor assigned (GET)	profile
GET /patients/{id}/emergency-card	patient own	minimal card for cache
GET · POST · PUT · DELETE /patients/{id}/contacts[/{cid}]	patient own (write), doctor (read)	trusted contacts
POST /patients/{id}/measurements	patient own; simulator in dev	save reading → evaluation (+ alert)
GET /patients/{id}/measurements?from&to&limit	patient own, doctor assigned	history
GET /patients/{id}/summary	patient own, doctor assigned	7 / 30-day stats from analysis
POST /patients/{id}/symptom-reports	patient own	symptom check → evaluation (+ alert)
GET /patients/{id}/alerts?state=open|all	patient own, doctor assigned	alerts list
GET /alerts/{aid}	owner / assigned doctor	alert + events
POST /alerts/{aid}/transition	per state machine	{to_state, note} → alert; 409 if not allowed
POST /patients/{id}/incidents	patient own	start incident {trigger, alert_id?}
POST /incidents/{iid}/actions	patient own	append one or many actions (offline queue)
PATCH /incidents/{iid}	patient own	location snapshot, outcome
GET /patients/{id}/incidents · GET /incidents/{iid}	patient own, doctor assigned	incident report with timeline
GET · POST /patients/{id}/journal	patient own (POST), doctor (GET)	journal
POST /journal/{jid}/review	doctor assigned	mark reviewed
GET · POST /patients/{id}/notes	doctor (POST, full GET); patient (GET advice only)	follow-up notes
GET · PUT /patients/{id}/care-plan	doctor assigned (PUT)	target values
POST /dev/simulate · POST /dev/reset-demo	only if DEV_TOOLS=1	simulator, reset seed

11.1 Key payloads

```
POST /api/v1/patients/2/measurements
{ "systolic": 186, "diastolic": 118, "pulse": 92,
  "measured_at": "2026-10-30T13:05:00Z", "source": "simulated",
  "context": { "rested_5_min": true, "position": "sitting" } }

201 Created
{ "measurement": { "id": 341, "systolic": 186, "diastolic": 118, "pulse": 92,
                   "measured_at": "2026-10-30T13:05:00Z", "source": "simulated",
                   "validation_status": "valid" },
  "evaluation": { "level": "severe", "rule_id": "R-SEV-1",
                  "reason": "Systolic 186 is at or above 180.",
                  "next_step": "Answer the symptom check, then rest and measure again in 60 seconds.",
                  "requires_symptom_check": true, "recheck_after_seconds": 60,
                  "show_emergency": false },
  "alert": { "id": 77, "level": "severe", "state": "new", "persistent": false } }
```

```
POST /api/v1/patients/2/symptom-reports
{ "symptoms": ["chest_pain"], "none_reported": false, "measurement_id": 341 }

201 Created
{ "report": { "id": 19, ... },
  "evaluation": { "level": "emergency", "rule_id": "R-EMG-1",
                  "reason": "Chest pain reported.",
                  "next_step": "Call emergency services now.",
                  "show_emergency": true },
  "alert": { "id": 77, "level": "emergency", "state": "new" } }   // level raised on alert 77
```

```
POST /api/v1/incidents/12/actions
{ "actions": [ { "action": "call_emergency_tapped", "client_time": "2026-10-30T13:06:10Z",
                 "detail": { "number": "14", "demo_mode": true } },
               { "action": "location_granted", "client_time": "2026-10-30T13:06:12Z",
                 "detail": { "accuracy_m": 24 } } ] }

Error format (all endpoints)
{ "error": { "code": "invalid_transition", "message": "Severe alerts cannot be dismissed.",
             "details": { "from": "new", "to": "dismissed" } } }
400 bad input · 401 no/expired token · 403 not your patient · 404 · 409 state conflict · 422 validation
```

12. Permissions

Data	Patient (self)	Assigned doctor	Other doctor / other patient	Trusted contact
Profile, contacts	read + write	read	none (403)	none (no account)
Measurements, summary	read + create	read	none	none
Alerts	read; acknowledge / dismiss notice	read; review / close	none	none
Incidents	create, add actions, outcome, read	read	none	only what the patient shares
Journal	create + read	read; mark reviewed	none	none
Doctor notes	read advice_for_patient only	create + read all	none	none
Care plan	read	read + write	none	none

Implemented in one place: `permissions.py` with `require_patient_access(user, patient_id, write=False)`. Every router uses it. `test_api_permissions.py` checks each route with a foreign patient and an unassigned doctor.

13. Data / AI module contract

Owned by the data/AI developer. Pure Python, no database, no network, fully unit-tested. The backend imports these functions only.

```
# api/app/analysis/validation.py
def validate_measurement(m: MeasurementIn, now: datetime) -> ValidationResult
    # -> ok: bool, status: "valid"|"questionable", errors: list[FieldError], notes: list[str]

# api/app/analysis/rules.py
def evaluate_measurement(m: Measurement, target: Target,
                         recent: list[Measurement], open_alerts: list[AlertView],
                         now: datetime) -> Evaluation
def evaluate_symptoms(symptoms: list[str], latest: Measurement | None,
                      open_alerts: list[AlertView]) -> Evaluation
    # Evaluation: level, rule_id, reason, next_step, requires_symptom_check,
    #             recheck_after_seconds, show_emergency,
    #             alert_action: "create"|"attach"|"raise"|"none", target_alert_id

# api/app/analysis/summary.py
def compute_summary(ms: list[Measurement], target: Target, now: datetime) -> Summary
    # avg_7d, avg_30d, count_7d, pct_above_target_7d, days_with_reading_7d,
    # trend: "rising"|"stable"|"falling"|"insufficient_data", last_measured_at, max_7d

# api/app/analysis/simulator.py
def scenario(name: str, start: datetime) -> list[MeasurementIn]   # "stable", "rising", "crisis"
```

13.1 Trend direction

Compare the mean of the last 7 days with the mean of the 7 days before. Rising if systolic mean is up by 5 mmHg or more, falling if down by 5 or more, otherwise stable. Fewer than 3 readings in either window gives `insufficient_data`. Simple and explainable on stage.

13.2 Seed data

- 1 doctor (Dr. Amina S.), 3 fictional patients assigned to her, 1 unassigned patient for permission tests.
- Karim B., 58: 30 days of readings drifting from about 135/85 to 150/95 (the demo patient). Trusted contact: his daughter (team phone in demo mode).
- Nadia M., 46: stable near target. Youssef T., 67: a severe reading 3 days ago, reviewed and closed with a note (shows history).
- All names, numbers and addresses fictional. reset-demo restores this state in under 5 seconds.

13.3 Where AI fits (decision D8)

Recommendation

- The safety-critical path (validation, alerts, emergency) is deterministic rules. No machine learning or LLM decides anything there.
- If time permits: an AI-drafted weekly summary **for the doctor only**, generated from computed statistics and journal entries, labelled 'AI draft, verify against data', shown under the chart. Never shown to the patient, never gives advice.
- In the pitch, call the rules 'explainable analysis' and the summary 'AI-assisted review'. Do not claim AI detection or diagnosis.

14. Screens

14.1 Patient (mobile first, French; emergency also in Arabic)

Screen	Content	Status
Header (all screens)	PULSE logo, red 'Urgence' button, DEMO MODE banner when on	MVP
Home	Last reading with colour and source tag, time since last reading, 7-day average vs target, open alerts, 'Add reading', 'I feel unwell'	MVP
Add reading	Systolic, diastolic, pulse, 'rested 5 min' checkbox, position. Result card with reason and next step. Severe → symptom check → 60 s timer → recheck	MVP
Symptom check	7 large checkboxes + 'None of these'. Any symptom → Emergency screen	MVP
History	Chart (target line, severe markers, source shapes) + list	MVP
Alerts	Open and past alerts, reason, state, acknowledge / dismiss (notice only)	MVP
Emergency	Section 7.3	MVP
Journal	Category, text, 'ask my doctor'; list with 'reviewed' and doctor's advice	MVP
Profile	Health info, regular doctor, trusted contacts, emergency card fields, location consent	MVP
Emergency setup	Medical ID guide, lock-screen card generator	If time
Weekly guidance	General cards: activity, salt, sleep; 'general information, not a prescription'	If time

14.2 Doctor (laptop first)

Screen	Content	Status
Patient list	Assigned patients sorted: open emergency, open severe, trend, then oldest last reading. Badges, last reading, 7-day average	MVP
Patient detail	Profile summary, chart with target and incidents marked, summary stats, open alerts, journal, notes	MVP
Alert review	Reason, rule, events timeline, buttons Reviewed / Close with note	MVP
Incident report	Trigger, timeline of actions with honest labels, location (if shared), outcome, readings around it, note form	MVP
Care plan	Set target systolic / diastolic with note	MVP
AI weekly summary	Draft panel, labelled	If time

15. Test plan and acceptance criteria

15.1 Rule test vectors (Data dev writes these first, before the rules)

#	Input	Expected
T1	128/82, target 140/90	no alert
T2	146/88, target 140/90	notice, R-TGT-1
T3	Second above-target reading while the notice is still open	attach to the open notice, no new alert
T4	4 readings in 7 days, mean 148/93, target 140/90	trend, R-TRD-1
T5	2 readings in 7 days, both high	no trend alert (not enough data)
T6	180/100	severe (boundary, ≥ 180)

#	Input	Expected
T7	160/120	severe (diastolic boundary)
T8	179/119	notice, not severe
T9	Severe, then recheck 184/121 after 70 s, no symptoms	severe persistent, R-SEV-2
T10	Severe, then recheck 165/100	severe stays open, R-SEV-3
T11	Symptom chest_pain, reading 130/80	emergency, R-EMG-1
T12	Severe alert open, then symptom vision_change	level raised to emergency on same alert
T13	120/130	reject (V2)
T14	310/100 or 'abc'	reject (V1)
T15	200/190	questionable (V4) and still severe
T16	measured_at 1 hour in the future	reject (V3)

15.2 API and integration tests (Ayoub)

- Permissions: each patient route called by another patient → 403; by unassigned doctor → 403.
- Flow: create reading → alert exists → doctor list shows patient first → transition review/close → events recorded.
- State machine: every allowed transition passes; dismissing a severe alert → 409; closing without a note on severe → 422.
- Incident: actions batch accepted twice with same client_time does not duplicate (idempotency on incident + action + client_time).
- reset-demo returns the exact seed counts.

15.3 Manual device checklist (Frontend, real phones, before hour 30)

- Android Chrome and iPhone Safari: tel: opens dialer with the number; share sheet opens; GPS prompt appears over HTTPS.
- Location denied → manual field shown, call button still first. Airplane mode → emergency screen still opens with card and numbers.
- Emergency screen reachable in one tap from every patient screen; text readable at arm's length.
- Simulated readings show the 'Simulated' tag in patient and doctor views.

15.4 Acceptance criteria (the demo is 'done' when all pass)

#	Criterion	Proved by
A1	Reading entered by the patient persists after refresh	Manual + test_api_flow
A2	Same reading appears in the assigned doctor's view within 15 s	Manual
A3	Rules are deterministic and covered by T1 to T16	pytest
A4	A red-flag symptom opens the emergency screen even with normal BP	T11 + manual
A5	Emergency call reachable in ≤ 2 taps from any patient screen; works with GPS off and offline	Checklist 15.3
A6	Emergency services, trusted contact and regular doctor are visibly separate	Review of screens
A7	Alerts, incidents and doctor notes stay in history after closing	test_state_machine
A8	No user reads another patient's data	test_api_permissions
A9	Simulated data is labelled; no unbuilt integration is claimed in UI or pitch	Review + demo script
A10	Full demo repeats from reset-demo three times in a row	Rehearsal

16. Demo script (about 4 minutes)

#	Where	What happens	Say
1	Doctor laptop	Dr. Amina's list: 3 patients; Karim's 30-day chart is rising (seeded).	'Between visits, doctors see nothing. PULSE fixes that.'
2	Patient phone	Karim enters 152/94 → notice with reason and next step.	'Every alert says why, and what to do.'
3	Simulator	Simulated reading 186/118 → severe → symptom check: none → 60 s timer.	'Labelled simulated: we did not have a connected cuff.'
4	Patient phone	Recheck 184/121 → persistent → 'contact your doctor today'.	'We follow published AHA patient guidance, pending clinician review.'
5	Patient phone	Karim reports chest pain → Emergency screen. Tap call (DEMO MODE rings a teammate). Show card, location, share to daughter.	'One button. The call is placed by the phone, not by us.'
6	Patient phone	Outcome: 'went to hospital'.	
7	Doctor laptop	Karim jumps to top. Incident report: timeline 'call tapped (not confirmed)', location, readings around it. Doctor writes note + follow-up date.	'His own doctor follows up, with the full record.'
8	Slide	Locked-phone answer: native Medical ID + lock-screen card.	'We are honest about what a web app can do.'

Backup: a recorded video of the full demo, made at hour 32, in case the venue network fails.

17. 36-hour plan

Hours	Ayoub (backend + merge)	Frontend dev	Data / AI dev
0 – 3	Approve decisions. Repo, CI, CODEOWNERS. schemas.py + stub routers + openapi.json (contract freeze).	Vite + Tailwind scaffold, routing, layout, header with Urgence button, static emergency screen from config.	Test vectors T1–T16 as failing tests. validation.py.
3 – 8	Models, DB, auth, permissions helper, measurements + symptom endpoints calling analysis, seed loader.	Login, Home, Add reading, result card, History (fixtures, then real API at ~h4).	rules.py passing all vectors. Seed dataset. Simulator script.
Checkpoint M1 (h8): Merge all. Vertical slice: patient adds reading → alert → doctor list shows it.			
8 – 16	Alert transitions + events, incidents + actions (idempotent), emergency-card + config endpoints, permission tests.	Symptom check, recheck timer, Alerts screen, Emergency screen live: tel, GPS, manual place, share, offline cache, action queue.	summary.py + trend; review chart correctness; texts_fr/ar.json for clinician review.
Checkpoint M2 (h16): Merge. Emergency flow end to end, persisted, visible to doctor.			
16 – 24	Doctor endpoints: patient list sort, notes, care plan, journal review. reset-demo. First deploy over HTTPS.	Doctor list, patient detail, alert review, incident report, notes, journal (patient + doctor).	Journal categories. If time: guidance.json, ai_summary.py spike behind a flag.
Checkpoint M3 (h24): MVP feature freeze. Only fixes and 'if time' items after this.			
24 – 30	Bug bash, edge cases, logs, README status table.	Real-phone checklist 15.3, mobile polish, if time: lock-screen card, Medical ID guide.	Re-run all vectors, check demo data story, if time: AI summary panel data.
30 – 33	Final deploy, demo mode on, number confirmation recorded, backup video.	Demo device setup (location permission granted, cache warm).	Pitch numbers and charts from seed.

Hours	Ayoub (backend + merge)	Frontend dev	Data / AI dev
33 – 36	Merge freeze at h34 (fixes only, Ayoub approves). Rehearse x3 from reset-demo.	Rehearse.	Rehearse.

Rest in turns: nobody codes more than about 8 hours without a break; the person resting must not hold an unmerged branch.

18. Risks and mitigations

Risk	Impact	Mitigation
GPS does not work on the phone (HTTP, permission, indoors)	Location missing in demo	Deploy over HTTPS; grant permission before demo; manual and home-address fallback always shown.
Dialing a real emergency number during tests or demo	Serious	Demo mode by default in dev; real number only behind config; never test-call.
Contract drift between frontend and backend	Integration breaks late	Contract freeze at h3, generated types, CI diff check, checkpoints.
Merge conflicts close to the deadline	Lost hours	Small branches, merge every ~3 h, freeze at h34, CODEOWNERS split by folder.
Scope creep (guidance, AI, maps)	MVP unfinished	Section 4 statuses; 'if time' work only after M3.
Unsafe or overconfident medical wording	Credibility and safety	All strings in one reviewed file; wording rules 5.4; mentor review.
Jury asks 'what if the phone is locked?'	Weak answer	Section 7.2 slide.
Venue network fails	Demo fails	Offline emergency screen; local server + tunnel; backup video.
Simulated data presented as real	Credibility	Source tag everywhere; say it in the pitch.
Wrong time display	Confusing timelines	UTC storage, Africa/Algiers display, one time helper.

19. Open decisions (team must settle before coding)

ID	Decision	Recommendation	Blocks coding?
D1	Stack	FastAPI + SQLite + React/Vite/TS (section 8)	Yes
D2	Default target before a doctor sets one	140/90; clinician to confirm (135/85 is the usual home-reading alternative)	No (config value)
D3	Number on the main emergency button	Protection Civile, confirmed by the mentor; 14 or 1021	No for coding, yes for demo
D4	Red-flag symptom list	AHA 7 symptoms	No
D5	Hosting	Render or Railway, one service; backup laptop + Cloudflare tunnel	No (by h16)
D6	Language	French UI; emergency screen French + Arabic	No
D7	Lock-screen card in or out	First 'if time' item after M3	No
D8	AI component	Doctor-only weekly summary, if time, labelled	No
D9	Demo phone	Android with Chrome as primary; test iPhone too	No (by h24)

ID	Decision	Recommendation	Blocks coding?
D10	Real BP device available?	Only use if already owned and tested by h12; otherwise simulated	No
D11	Repo host and name	GitHub, private, 'pulse', Ayoub owner	Yes

Questions only the team can answer

- Is there a medical mentor at the hackathon who can review section 5 and the emergency number? (Strongly recommended.)
- What is the frontend developer's strongest framework? If not React, D1 should follow their strength for the frontend only.
- Does the data/AI developer prefer Python? If not, D1 changes.
- Does the jury require a working deployed URL, or is a local demo accepted?

Review checklist for the team

- Is the priority order in section 1 right?
- Is anything marked MVP in section 4 that should be 'if time', or the reverse?
- Are the rules in section 5 understandable enough to explain on stage in 30 seconds?
- Can each member build their part of section 17 in the hours given?
- Is any claim in the demo script (section 16) not backed by something we will actually build?

20. Sources

Checked on 29 October 2026.

- American Heart Association, 'Hypertensive Crisis: When You Should Call 911 for High Blood Pressure' (threshold 180/120 or higher, recheck after 1 to 2 minutes, symptom list). goredforwomen.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings/hypertensive-crisis-when-you-should-call-911-for-high-blood-pressure (the heart.org page with the same content could not be loaded at the time of checking).
- UK Foreign, Commonwealth and Development Office, 'Algeria: getting help' (ambulance 1021, fire 14, police 17 or 1548). gov.uk/foreign-travel-advice/algeria/getting-help
- Wikipedia, 'Telephone numbers in Algeria' (police 1548, gendarmerie 1055, civil defence 1021). en.wikipedia.org/wiki/Telephone_numbers_in_Algeria
- Referenced by the team handoff, not re-checked: Algeria Civil Protection, dgpc.dz; US Department of State Algeria page.
- Team inputs: hackathon problem statement 'Problématique 2', first team concept notes, PULSE conception handoff and report (29 October 2026).

Final note

This is a hackathon prototype specification, not medical advice and not a validated clinical protocol. Thresholds, symptom lists and all patient-facing text need review by a qualified clinician before real use.